Vision Aided Navigation - Exercise 1

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GitHub Repository: https://github.com/ShayMorad/3D-Computer-Vision-SLAM

(Note: Code is located on the main branch. Collaborator invites have been sent to course teachers).

Part 1.1: Detect and Extract Key-points

The key-points were detected using SIFT.

- **Left Image Key-points:** 1551
- **Right Image Key-points:** 1661

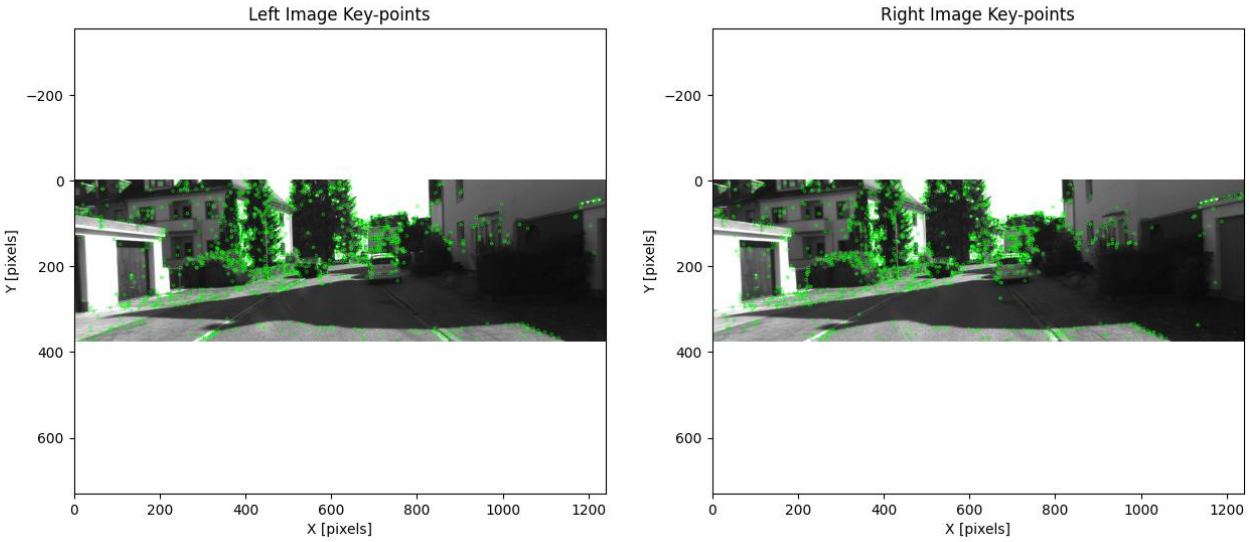


Figure 1: Detected SIFT key-points.

Code Reference: ex1.py, Line 33

Part 1.2: Feature Descriptors

Below are the calculated descriptors for the first feature of the left image and the first feature of the right image:

Descriptor of the first feature (Left Image):

[3. 14. 28. 20. 3. 16. 23. 12. 94. 4. 11. 28. 4. 11.
30. 61. 216. 3. 1. 1. 0. 0. 1. 148. 79. 10. 6. 4.
3. 0. 0. 15. 5. 23. 45. 31. 4. 10. 13. 8. 185. 39.
38. 16. 1. 5. 13. 21. 216. 22. 1. 1. 2. 0. 0. 53.
66. 5. 2. 14. 13. 0. 0. 6. 4. 21. 29. 10. 0. 0.
0. 0. 150. 16. 16. 7. 0. 0. 0. 3. 216. 5. 0. 0.
2. 0. 0. 25. 31. 0. 0. 2. 11. 1. 0. 5. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
0. 0.]

Descriptor of the first feature (Right Image):

[92. 1. 0. 2. 1. 0. 0. 29. 15. 2. 11. 53. 9. 0.
0. 2. 15. 12. 15. 16. 0. 0. 0. 1. 4. 0. 0. 0.
0. 0. 0. 1. 105. 1. 0. 9. 18. 0. 0. 33. 34. 19.
29. 165. 161. 0. 0. 1. 165. 65. 36. 43. 14. 0. 0. 17.
43. 1. 0. 0. 0. 0. 0. 9. 75. 1. 0. 3. 42. 7.
0. 26. 66. 5. 1. 35. 165. 53. 0. 5. 165. 14. 1. 7.
38. 9. 0. 20. 44. 0. 0. 0. 0. 0. 0. 5. 61. 1.
0. 2. 27. 6. 0. 25. 54. 4. 1. 40. 165. 40. 0. 4.
165. 8. 0. 10. 39. 5. 0. 23. 28. 0. 0. 0. 0. 0.
0. 4.]

Code Reference: ex1.py, Line 28

Part 1.3: Match Descriptors

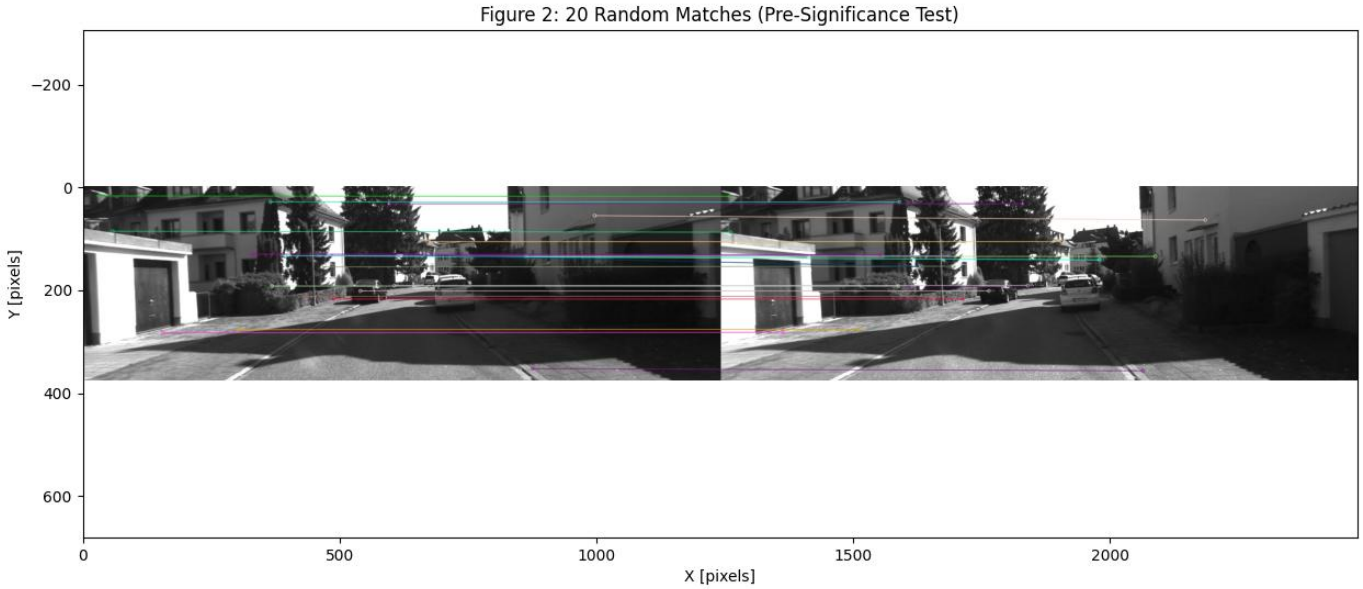


Figure 2: 20 Random Descriptor Matches.

Code Reference: ex1.py, Line 46

Part 1.4: Significance Test

- **Ratio value used:** 0.75
- **Matches discarded:** 795

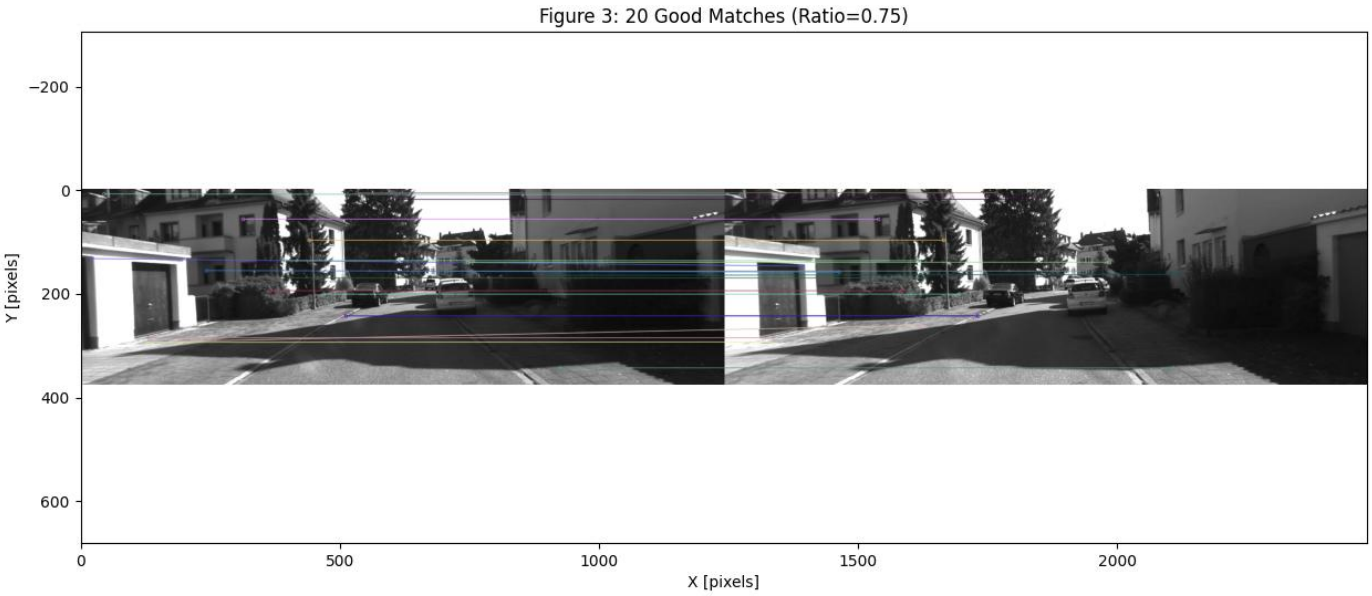


Figure 3: 20 Matches Post-Significance Test.

Code Reference: ex1.py, Line 60

Correct Match That Failed the Test:

- **Ratio value used: 0.4**



Figure 4: A geometrically correct match rejected by the significance test.

Code Reference: ex1.py, Line 78

Explanation: This geometrically correct match was discarded because it was located in an area with a highly repetitive visual pattern. Because the feature pattern repeats locally, both the 1st nearest neighbor and the 2nd nearest neighbor yielded highly similar descriptor distances. This resulted in a distance ratio close to 1.0, thereby failing the strict distance ratio test, even though the spatial match was correct.